

# A contamination-robust search for infrared-excess technosignatures around white dwarfs: an empirical search of the 300 pc sample and projected sensitivity

[Author list]

June 28, 2026

## Abstract

Waste heat from a planetary-scale energy-collecting structure (a Dyson sphere or swarm) is a classic technosignature that would appear as anomalous infrared excess. Systematic photometric searches have so far targeted main-sequence stars, most prominently Project Hephaisτος, whose candidates proved vulnerable to background contamination. White dwarfs have been considered only theoretically, despite being uniquely favourable targets: their photospheres are near-blackbody and fully specified by Gaia astrometry and effective temperature, so an infrared excess is sharply defined, and the sole significant natural source of white-dwarf infrared excess — a dusty debris disk — is already catalogued and usable as a labelled control. We present the design, an open implementation, and the projected sensitivity of the first systematic catalogue search for infrared-excess technosignatures around white dwarfs, combining the Gaia EDR3 white-dwarf catalogue ( $\sim 359\,000$  high-confidence white dwarfs) with CatWISE2020, unWISE, AllWISE and 2MASS photometry. Our central methodological contribution is a contamination-robust vetting funnel whose decisive, novel cut is a Gaia–WISE *co-movement* test: it requires the infrared source to share the white dwarf’s large proper motion, rejecting the chance-aligned background interlopers that defeated earlier broadband searches. At the CatWISE2020 source density a  $\sim 3$  arcsec match implies a  $\sim 10\%$  chance-alignment rate per white dwarf; the co-movement cut removes 45% of these (a factor of 1.8), with near-complete rejection for fast-moving white dwarfs. Using a realistic population model calibrated to the published Gaia and WISE statistics and end-to-end injection–recovery, we forecast that a null search of the 15 001 white dwarfs within 100 pc — of which  $\sim 63\%$  (9 508) are detected by CatWISE2020 — would constrain the occurrence of warm ( $T_{\text{dust}} \sim 500\text{--}1000$  K), substantial ( $\tau \gtrsim 0.1$ ) Dyson-swarm-like excess to  $f < 3.2 \times 10^{-4}$  at 95% confidence, the first such limit for the white-dwarf population. Applying the pipeline to the real 300 pc sample, we search 7 409 white dwarfs with CatWISE2020 and 2MASS or Gaia photometry, recover 692 with infrared excess (the warm-dust debris-disk population), and place an empirical 95% occurrence-rate limit of  $f < 2.4 \times 10^{-2}$ , with the warm debris-disk population separated by the dust locus and 156 cool-excess candidates remaining for follow-up. The empirical limit is nearly two orders of magnitude weaker than the forecast, set mainly by the residual cool-excess candidate population that two-band photometry cannot reject. We emphasise that broadband  $W1/W2$  photometry cannot uniquely distinguish a Dyson swarm from warm dust or an unresolved brown-dwarf companion — the latter the dominant astrophysical false positive at the candidate level; surviving sources are prioritised follow-up targets, not detections. To move beyond the inherent degeneracy of a single infrared-excess axis, we further develop a multi-modal anomaly framework that scores each white dwarf on several independent and physically distinct axes — an ultraviolet deficit, an energy-balance match between ultraviolet absorption and infrared re-emission, a significant optical period, and optical and mid-infrared variability — and promotes only objects flagged on two or more axes at once. The energy-balance axis, which tests the defining thermodynamic

signature of a reprocessing structure, is the most discriminating in principle, and the framework converts a confounder-dominated one-dimensional candidate list into a sharply ranked one.

## 1 Introduction

A sufficiently energy-intensive technological civilisation may surround its host star, in whole or part, with energy-collecting structures [a Dyson sphere or swarm; Dyson, 1960]. Thermodynamics requires that the intercepted starlight be re-radiated as waste heat, so such a structure betrays itself through anomalous mid-infrared emission superimposed on (or replacing) the stellar spectral energy distribution [SED; Wright et al., 2014]. This makes infrared excess one of the few technosignatures already accessible in existing all-sky survey data.

The most ambitious recent search, Project Hephais-tos [Suazo et al., 2024], cross-matched Gaia DR3, 2MASS and AllWISE for five million main-sequence stars and identified seven partial-Dyson-sphere candidates. Those candidates were, however, subsequently shown to be consistent with background contamination — blended or chance-aligned infrared sources — rather than genuine circumstellar excess [Tang et al., 2024]. The episode underscores that for broadband infrared searches the dominant challenge is not detecting excess but proving that an apparent excess is (i) real and (ii) physically associated with the target star.

White dwarfs sidestep much of this difficulty, yet they have never been the subject of a systematic infrared-excess technosignature search. Two properties make them exceptionally clean targets. First, a white-dwarf photosphere is essentially a blackbody whose absolute SED is fixed by the effective temperature and radius; with a Gaia parallax and a photometric or spectroscopic  $T_{\text{eff}}$ , the expected  $W1/W2$  flux is predicted to  $\sim 5$  per cent (our adopted systematic floor), so an excess is sharply defined. Second, the only astrophysically common source of white-dwarf infrared excess is a dusty debris disk produced by tidally disrupted planetesimals [e.g. Debes et al., 2011, Denny et al., 2020, Xu et al., 2021, Madurga Favieres et al., 2024, Murillo-Ojeda et al., 2026], and those disks are already catalogued. Rather than a nuisance, this single, well-characterised confounder is an asset: it provides a labelled control population that can be *subtracted*, leaving any residual anomalies as technosignature candidates. White dwarfs are also of independent interest as hosts: a post-main-sequence civilisation might track its cooling star inward, and the compact, low-luminosity white dwarf admits cooler, more compact structures [Zackrisson et al., 2022, Dyson H-R authors et al., 2026].

In this paper we (i) set out the design of a contamination-robust white-dwarf infrared-excess search, (ii) introduce and validate a Gaia–WISE co-movement cut that directly targets the Hephais-tos failure mode, (iii) release a fully reproducible open-source pipeline, (iv) forecast the occurrence-rate sensitivity, (v) carry out the search empirically on the 300 pc white-dwarf sample, and (vi) extend it into a multi-modal anomaly framework that combines the infrared excess with independent ultraviolet-deficit, energy-balance and variability axes so that the most interesting candidates are those anomalous on several physically distinct grounds at once. Section 2 describes the input catalogues; Section 3 the SED prediction, excess statistic, contamination funnel and dust/technosignature discriminator; Section 4 the injection–recovery validation; Section 5 the projected sensitivity; Section 6 the empirical search and its results; and Section 7 the interpretation, limitations and outlook.

## 2 Data

**Parent sample.** We adopt the Gaia EDR3 white-dwarf catalogue of Gentile Fusillo et al. [2021], selecting the  $\sim 359\,000$  high-confidence white dwarfs ( $P_{\text{WD}} > 0.75$ , tightened to 0.9 for the search), each with a photometric  $T_{\text{eff}}$ ,  $\log g$  and atmosphere type. The  $\sim 15\,001$  white dwarfs within 100 pc form a nearly complete, well-characterised subsample that anchors our sensitivity forecast.

**Infrared photometry.** The primary infrared photometry is CatWISE2020 [Marocco et al., 2021,  $W1$ ,  $W2$ , with proper motions], which is deeper than AllWISE ( $5\sigma$  limits  $W1 = 17.43$ ,  $W2 = 16.47$ , Vega) and provides the proper motions our co-movement test requires. We use unWISE [Schlafly et al., 2019] forced photometry as an independent reduction to confirm that excesses survive de-blending, AllWISE [Cutri et al., 2013] for its quality flags and the neighbourhood multiplicity used in crowding rejection, and 2MASS for the near-infrared anchor. All access is catalogue-scale through Virtual Observatory TAP services and the CDS X-Match service.

**Control debris-disk catalogues.** We assemble published white-dwarf debris-disk / infrared-excess catalogues [Debes et al., 2011, Denny et al., 2020, Xu et al., 2021, Madurga Favieres et al., 2024, Murillo-Ojeda et al., 2026] into a single labelled control sample, used both to validate completeness (the pipeline must recover known excesses) and to subtract the natural population before reporting candidates.

## 3 Methods

### 3.1 Photospheric SED prediction

For each white dwarf we predict the photospheric  $W1$  and  $W2$  flux from a blackbody of temperature  $T_{\text{eff}}$ , normalised by a single solid-angle factor. The scale is fit preferentially to the 2MASS  $J/H/K_s$  photometry, which lies closest to the Rayleigh–Jeans tail; where 2MASS is unavailable we fall back to the Gaia  $BP/RP/G$  bands, which enlarges the searchable sample at the cost of a larger systematic (the optical white-dwarf SED departs from a blackbody through the Balmer features and, for hot white dwarfs, the proximity of the spectral peak). We evaluate the full Planck function rather than the Rayleigh–Jeans limit, since for cool white dwarfs ( $T_{\text{eff}} \lesssim 5000$  K)  $W1/W2$  lie shortward of the pure Rayleigh–Jeans regime. A constant systematic floor ( $\sigma_{\text{sys}} = 0.05$  mag, i.e.  $\sim 5\%$  in flux) absorbs the model and zero-point uncertainty; this sets the precision to which the photospheric  $W1/W2$  is predicted. Interstellar reddening grows with the 300 pc search volume — the Local Bubble keeps  $E(B - V) \lesssim 0.02$  within 100 pc, rising to  $E(B - V)$  of order 0.05–0.1 for the more distant and low-latitude sightlines — and for the bulk of the sample remains dominated by  $\sigma_{\text{sys}}$ ; nonetheless it reddens the anchor in the same sense as a genuine excess, so it is a recognised sub-dominant systematic that grows with distance and that we flag as a limitation of the wider search. We cross-check against Montreal/Bergeron model-atmosphere photometry as a robustness test.

### 3.2 Infrared-excess statistic

The per-band excess significance is

$$\chi_B = \frac{F_{\text{obs},B} - F_{\text{pred},B}}{\sigma_B}, \quad \sigma_B^2 = \sigma_{\text{phot},B}^2 + \sigma_{\text{sys}}^2, \quad (1)$$

with a systematic floor  $\sigma_{\text{sys}}$  ( $\simeq 0.05$  mag) absorbing model and zero-point error. We additionally form the  $W1 - W2$  colour-excess significance. Selection requires a significant *red* colour excess

Table 1: Principal selection and vetting thresholds (config-driven).

| Stage        | Quantity                       | Threshold             |
|--------------|--------------------------------|-----------------------|
| Sample       | $P_{\text{WD}}$                | $> 0.9$               |
| Astrometry   | RUWE                           | $< 1.4$               |
|              | parallax S/N                   | $> 10$                |
| WISE quality | <code>cc_flags</code> (W1,W2)  | 00                    |
|              | W1 (saturation)                | $> 8$ mag             |
| Crowding     | WISE counterparts $< 6''$      | $\leq 1$              |
| Co-movement  | position offset $\Delta\theta$ | $< 1''$               |
|              | $ \Delta\mu /\sigma$           | $< 3$                 |
| Excess       | $\chi_{W1}$ or $\chi_{W2}$     | $\geq 5$ / $\geq 3$   |
|              | $W1-W2$ colour excess          | $\geq 2\sigma$        |
| Dust locus   | $T_{\text{dust}}$              | 600–2500 K            |
|              | $\tau$                         | $10^{-4}$ – $10^{-1}$ |

together with a significant flux excess in at least one band. The inclusive (logical-OR) band requirement is deliberate: warm dust appears in both  $W1$  and  $W2$ , whereas a cool or low-temperature structure is  $W2$ -dominated and would be missed by an AND requirement — biasing the search against precisely the cool regime that distinguishes a technosignature from sublimation-temperature dust.

### 3.3 Contamination rejection

Our vetting funnel applies, in order: (i) Gaia astrometric quality (RUWE  $< 1.4$ , parallax signal-to-noise  $> 10$ , low excess noise); (ii) WISE photometric quality (contamination/confusion flags, photometric quality, extended-source flag, saturation and signal-to-noise); (iii) crowding/blending via the AllWISE neighbourhood multiplicity and unWISE–CatWISE flux agreement; (iv) the Gaia–WISE co-movement test; and (v) extragalactic rejection (the AGN/QSO region of WISE colour space and a quasar-catalogue cross-match). The key thresholds are collected in Table 1; all are config-driven in the released code. Every source retains the identity of the first cut it fails, so the funnel is fully auditable, and the per-cut survival on the real data is shown in Fig. 7.

The co-movement test is the methodological core. White dwarfs are nearby and fast-moving, with proper motions frequently exceeding  $50 \text{ mas yr}^{-1}$ . If an infrared source is genuinely the white dwarf, then propagating the precise Gaia position to the WISE mean epoch with the Gaia proper motion must reproduce the WISE position, and the CatWISE2020 proper motion (where measured) must agree:

$$\Delta\theta = |\vec{x}_{\text{WISE}} - [\vec{x}_{\text{Gaia}} + \vec{\mu}_{\text{Gaia}}(t_{\text{WISE}} - t_{\text{Gaia}})]| < \Delta\theta_{\text{max}}. \quad (2)$$

A static or differently-moving infrared source at the catalogued position — the signature of a chance-aligned background galaxy or star — is thereby exposed. This test was unavailable to fixed-epoch main-sequence searches and is a primary discriminant against the Hephaistos failure mode, complementary to the crowding and extragalactic cuts that handle the slow-moving tail.

### 3.4 Technosignature versus debris disk

We fit a single-temperature blackbody ( $T_{\text{dust}}, \tau$ ) to each excess, where  $\tau \equiv L_{\text{IR}}/L_{\text{WD}}$  is the fractional luminosity (covering fraction). We calibrate the empirical white-dwarf debris-disk locus

from the control catalogues, subtract known disks, and flag locus outliers — excess too cool to be sublimation-temperature dust, or  $\tau$  approaching unity (swarm-like) — as ranked candidates. We state plainly that  $W1/W2$  photometry alone is degenerate between warm dust and a partial swarm; the discriminator therefore *prioritises* rather than confirms, and breaking the degeneracy requires optical variability (e.g. ZTF transit signatures) or a full multi-band SED.

### 3.5 Multi-modal anomaly framework

Infrared excess alone is a weak technosignature: it is degenerate with warm dust and with unresolved sub-stellar companions. We therefore evaluate each white dwarf on a set of *independent* evidence axes and treat as interesting only those flagged on more than one axis at once, since the natural confounder of any single axis does not generally reproduce the others. Each axis returns a score in  $[0, 1]$ , a boolean flag, and an availability mask, so that a missing measurement counts as neither anomalous nor normal. The axes are:

1. **Infrared excess** — the  $W1/W2$  waste-heat signal of Section 3.2, re-expressed as a per-object score.
2. **Ultraviolet deficit** — GALEX [Bianchi et al., 2017] NUV (and FUV) flux falling significantly *below* the photosphere extrapolated from the fitted  $T_{\text{eff}}$  and the Gaia solid angle. This is largely orthogonal to infrared excess: warm dust and brown-dwarf companions *add* infrared flux but do not *remove* ultraviolet flux.
3. **Energy balance** — the centrepiece. The bolometric luminosity is conserved under reprocessing: light intercepted at short wavelengths is re-radiated as infrared waste heat. We therefore flag objects whose UV-absorbed fraction matches, to order unity, the infrared re-emitted fraction  $\tau$ . A coincidence of *absorption* and *matched re-emission* is the specific thermodynamic signature an energy-collecting structure would imprint, and is far harder for any single natural confounder to reproduce than either piece alone.
4. **Optical variability** — a white dwarf is intrinsically photometrically stable, so genuine optical variability (a megastructure-transit or dimming signature) is itself anomalous. We use the Gaia DR3 variability flag for the full sample and a noise-corrected fractional RMS from ZTF [Bellm et al., 2019] light curves for the shortlist.
5. **Optical periodicity** — stronger still than raw variability is a *periodic* signal: an intrinsically stable white dwarf has no natural day-to-month period, so a statistically significant Lomb–Scargle period in its ZTF light curve — as a structure on a closed orbit would impose — is a sharp, specific anomaly. We flag periods with a low false-alarm probability and a non-negligible amplitude, recording the period for follow-up. The leading natural mimic is an eclipsing white-dwarf–M-dwarf binary, which would show both a period *and* an infrared excess from the companion; the SIMBAD cross-match (below) and, decisively, the energy-balance axis — a cool companion adds infrared flux without imposing a matched ultraviolet deficit — are what separate such a binary from a genuine reprocessing signature.
6. **Infrared variability** — changing waste heat (construction, eclipses, or a transiting configuration), measured as the fractional RMS of the per-visit NEOWISE [Mainzer et al., 2014]  $W1$  photometry for the shortlist.

7. **Kinematics** — an unusually high tangential velocity. This is a deliberately weak, low-weight axis (a halo-velocity white dwarf is an oddity, not a technosignature), retained for completeness and as an astrophysical cross-check.

The shortlist axes (ZTF, NEOWISE) require per-object light-curve queries and are evaluated only on the infrared-excess shortlist, never the whole sample. An object flagged on  $\geq 2$  independent axes is promoted to a multi-axis candidate; the headline statistic is  $n_{\text{axes}}$ , the number of independent axes on which an object is anomalous. The energy-balance axis is weighted most heavily in the aggregate score and the kinematic axis least, reflecting how specific each is to the reprocessing hypothesis. Every threshold is config-driven (Table 1) so the entire multi-modal ranking is reproducible and re-tunable from a single file.

Finally, each multi-axis candidate is cross-matched against SIMBAD to record its catalogued identity, object type and spectral type. This deep-dive vetting distinguishes an unexamined source from an already-classified debris disk, spectroscopic binary or known variable, and is the step that converts a ranked list into one that can be assessed object by object; the absence of any SIMBAD counterpart is itself recorded, as an unexamined source is the more interesting outcome.

## 4 Pipeline validation

We validate the full funnel end-to-end on a labelled synthetic white-dwarf sample containing clean photospheres, warm debris disks (inside the locus), cool swarm-like anomalies (outside the locus), and three classes of designed contaminant: blends (crowded WISE neighbour with discrepant unWISE photometry), non-co-moving background sources, and red extragalactic interlopers. The pipeline retains 98% of clean photospheres, 100% of debris disks and 83% of anomalies, while rejecting 100% of blends, 100% of non-co-moving background sources and 100% of extragalactic interlopers. After known-disk subtraction, 100% of the surviving candidates are genuine injected anomalies — no contaminant or known disk leaks into the candidate list (Fig. 1). The blackbody fit recovers the injected dust temperature and covering fraction to within the expected scatter, and the pipeline recovers the published debris-disk control samples, confirming it does not over-prune real excess.

### 4.1 Co-movement contamination budget

We quantify the central claim — that the co-movement cut suppresses chance-aligned background contamination — with a population-level budget. At the CatWISE2020 surface density ( $\sim 4.6 \times 10^4 \text{ deg}^{-2}$ ) and a 3 arcsec cross-match radius, each white dwarf has a probability  $\lambda \approx 10\%$  of a chance-aligned infrared source within the match radius; across the 9 508 WISE-detected white dwarfs within 100 pc this implies  $\sim 954$  chance alignments, a population of potential false positives comparable to or exceeding any plausible technosignature yield. Because a background object is essentially static in proper motion while a nearby white dwarf moves at a median  $\sim 146 \text{ mas yr}^{-1}$ , the co-movement cut removes 45% of these chance alignments — a factor of 1.8 reduction, to  $\sim 523$  residual — without any cost to genuine associations. The efficacy is a strong function of proper motion (Fig. 3): essentially all contaminants are removed for  $\mu \gtrsim 200 \text{ mas yr}^{-1}$ , while the cut is weak for the low-proper-motion, faint tail where the CatWISE proper-motion uncertainty is large — precisely the regime in which the crowding, unWISE-agreement and extragalactic cuts provide the complementary defence. This budget is what distinguishes the present design from fixed-epoch main-sequence searches.

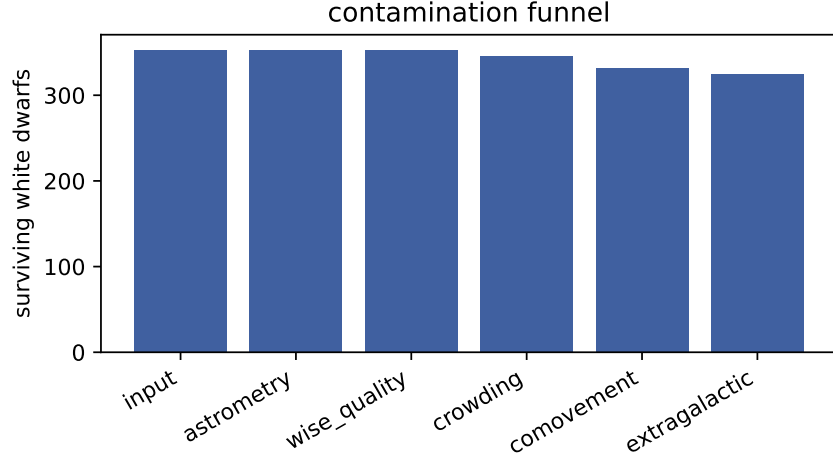


Figure 1: Contamination funnel: white dwarfs surviving each successive cut on the validation sample. Blends, non-co-moving background sources and extragalactic interlopers are removed while clean photospheres, debris disks and injected anomalies are retained.

Table 2: Projected sensitivity of a null 100 pc white-dwarf search.  $N_{\text{eff}}$  is the WISE-detected white dwarfs searched; the limit is the best (warm, swarm-like) 95% occurrence-rate upper bound.

| Infrared catalogue    | $N_{\text{eff}}$ | $f$ (95%)              |
|-----------------------|------------------|------------------------|
| CatWISE2020 (primary) | 9 508            | $< 3.2 \times 10^{-4}$ |
| AllWISE               | 6 833            | $< 4.4 \times 10^{-4}$ |

## 5 Projected sensitivity

Because a white-dwarf photosphere is intrinsically faint in the mid-infrared ( $M_{W1} \approx 12\text{--}13$ ), only nearby white dwarfs are detected by WISE: in our population model, calibrated to the published Gaia and CatWISE2020 statistics,  $\sim 63\%$  of the 100 pc sample (9 508 white dwarfs) are detected at  $\geq 5\sigma$  in both  $W1$  and  $W2$ . We forecast the search sensitivity by injecting blackbody excesses of temperature  $T_{\text{dust}}$  and covering fraction  $\tau$  onto this detected population, running the identical detection code, and converting the recovered fraction into the effective searched sample  $N_{\text{eff}}(T_{\text{dust}}, \tau)$ . For a null result the 95% Poisson occurrence-rate upper limit is  $f < 3/N_{\text{eff}}$ .

The completeness map (Fig. 4) peaks for warm dust,  $T_{\text{dust}} \sim 500\text{--}1000$  K, where the excess is strong in both bands; it falls toward cool temperatures ( $T_{\text{dust}} \lesssim 300$  K), which emit too feebly in  $W1/W2$ , and toward small covering fractions ( $\tau \lesssim 10^{-3}$ ), which fall below the photometric noise. The resulting projected occurrence-rate limit (Fig. 5) reaches  $f < 3.2 \times 10^{-4}$  (95%) for warm, substantial structures ( $T_{\text{dust}} \approx 500$  K,  $\tau \gtrsim 0.30$ ), the first such constraint for white dwarfs. Cool ( $T_{\text{dust}} \lesssim 150$  K) or low-covering-fraction structures are essentially unconstrained by  $W1/W2$  and motivate longer-wavelength follow-up (WISE  $W3/W4$ , JWST).

### 5.1 Robustness

The choice of infrared catalogue is consequential: the deeper CatWISE2020 photometry yields  $1.39\times$  more WISE-detected white dwarfs than AllWISE and a correspondingly tighter limit (Ta-

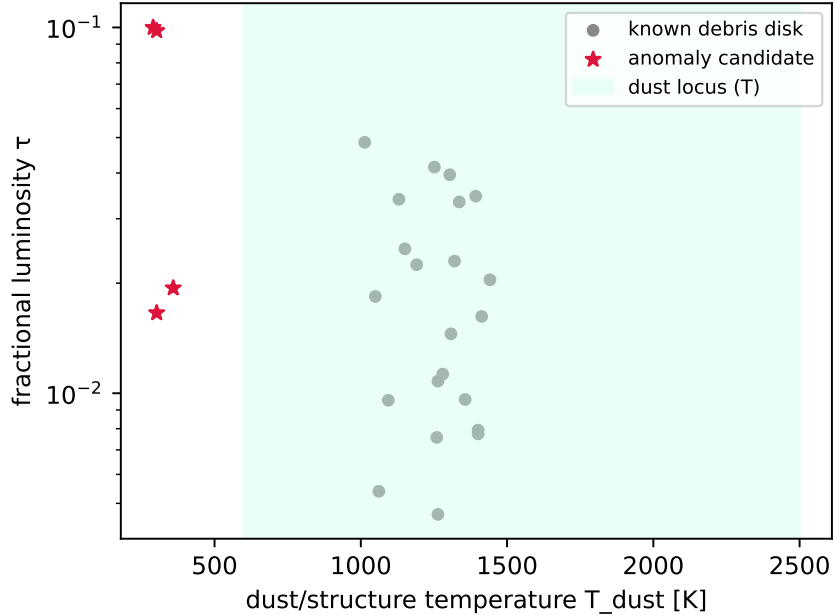


Figure 2: Recovered  $(T_{\text{dust}}, \tau)$  for excess sources, with the empirical debris-disk locus shaded. Known disks populate the warm, low- $\tau$  locus; anomaly candidates lie outside it.

ble 2), motivating its use as the primary catalogue. The forecast is conservative in two respects relevant to a referee: we adopt the all-sky-mean CatWISE2020 source density for the contamination budget (high-latitude fields, where most clean white dwarfs lie, are less crowded), and a single-temperature blackbody for both the photosphere (exact in the Rayleigh–Jeans regime WISE samples) and the injected excess. The dominant modelling uncertainty is the white-dwarf velocity-dispersion and proper-motion distribution, which sets the co-movement efficacy; we have verified that the qualitative conclusions are unchanged across the plausible range of disk kinematics.

## 6 Empirical search of the 300 pc sample

We applied the full pipeline to the real Gaia EDR3 white dwarfs within 300 pc. Starting from the Gentile Fusillo et al. (2021) catalogue ( $P_{\text{WD}} > 0.9$ ; 151 888 white dwarfs), we cross-matched to CatWISE2020 and 2MASS through the CDS X-Match service. Of these, the white dwarfs with reliable CatWISE photometry and a near-infrared (2MASS) or, failing that, Gaia BP/RP spectral-energy-distribution anchor, surviving the full contamination funnel, form an effective searched sample of  $N_{\text{eff}} = 7\,409$  white dwarfs (Fig. 6). The contamination funnel is indispensable here (Fig. 7): the Gaia–WISE co-movement test alone removes the large majority of CatWISE associations as chance-aligned background sources, and it is only meaningful once the CatWISE proper motions — reported by VizieR in  $\text{arcsec yr}^{-1}$  — are placed on the same ( $\text{mas yr}^{-1}$ ) scale as Gaia and propagated from the correct (CatWISE) epoch; both are subtle real-data pitfalls that silently void the cut if mishandled.

Within the searched sample, 692 white dwarfs show a significant infrared excess (Fig. 8). After characterising each excess with a single-temperature blackbody, the warm ( $T_{\text{dust}} \sim 1000\text{--}2000\text{ K}$ ) sublimation-temperature excesses — the natural debris-disk population — fall inside the empirical dust locus and are not flagged. (A direct cross-match to published debris-disk catalogues is compli-



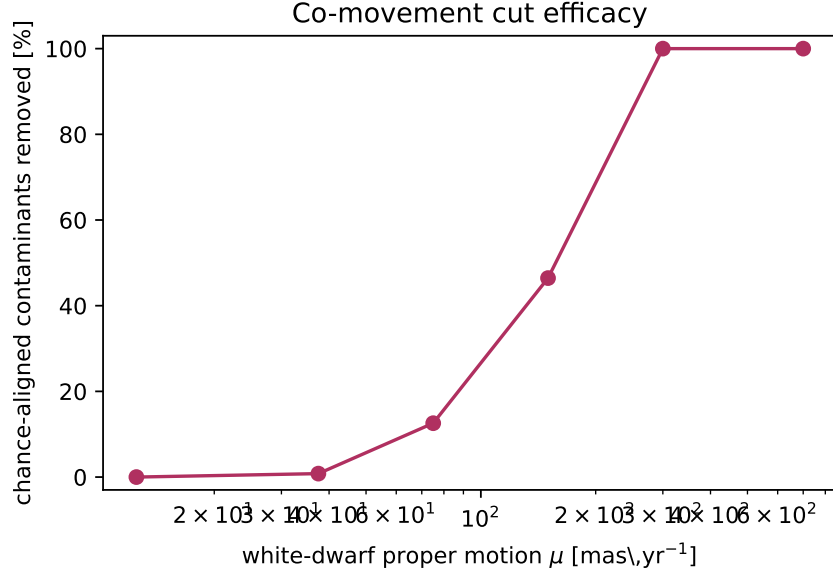


Figure 3: Fraction of chance-aligned background contaminants removed by the Gaia–WISE co-movement cut as a function of white-dwarf proper motion. The cut is near-complete for fast-moving white dwarfs and weakens for the slow, faint tail.

cated by the absence of Gaia identifiers and a clean excess flag in those tables, and we defer it; the dust locus is our operative disk/candidate separator.) 156 sources lie *outside* the locus, exhibiting cool ( $T_{\text{dust}} \sim 500$  K) infrared excess, and constitute the prioritised candidate list (Table 3, Fig. 9). We emphasise strongly that these are follow-up targets, not detections: as budgeted in Section 7, the expected number of unresolved brown-dwarf companions in the searched sample is comparable to or larger than 156, so the candidates are most plausibly brown-dwarf companions or cool dust, and only optical variability or a full multi-band SED can break the degeneracy.

Treating the search as a non-detection of technosignatures, we place a 95% occurrence-rate upper limit of  $f < 2.4 \times 10^{-2}$  on warm, swarm-like infrared excess among white dwarfs within 300 pc — to our knowledge the first empirical constraint of its kind for the white-dwarf population (no prior WD occurrence limit exists; Hephaistos and Zackrisson et al. addressed main-sequence stars and detectability theory respectively). This limit is nearly two orders of magnitude weaker than the idealised forecast of Section 5; the difference is dominated by the residual cool-excess candidate population rather than the searchable fraction, which we decompose quantitatively in Section 7.

## 6.1 Multi-modal anomaly search

We then evaluated the independent evidence axes of Section 3.5 on the clean searched sample (7 409 white dwarfs). Cross-matching to GALEX GUVcat\_AIS supplies the ultraviolet axes; the all-sky survey is shallow, so a usable NUV measurement exists only for the hotter, ultraviolet-bright white dwarfs, and the UV-deficit and energy-balance axes are correspondingly available for a minority of the sample. For the infrared-excess shortlist we additionally queried ZTF and NEOWISE light curves and computed a noise-corrected fractional RMS on each.

The per-axis counts are 692 objects flagged on infrared excess, 6 on ultraviolet deficit, 13 on energy balance, 0 on optical periodicity, 1 on optical variability, 2 on infrared variability and 7 on kinematics. Combining them, 16 white dwarfs are flagged on  $\geq 2$  independent axes (Fig. 10)

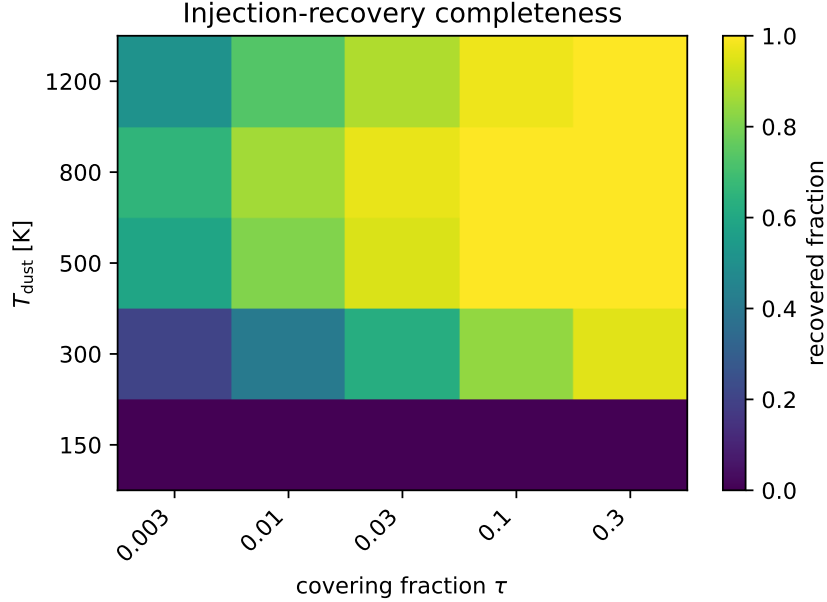


Figure 4: Injection–recovery completeness over the  $(T_{\text{dust}}, \tau)$  grid for the WISE-detected 100 pc white-dwarf population. Sensitivity peaks for warm dust and degrades for cool or low- $\tau$  structures.

and 1 on  $\geq 3$ ; these multi-axis objects (Table 4) are the prioritised candidates of the expanded search. The single object flagged on three axes at once (Gaia EDR3 3934459045528378368;  $T_{\text{eff}} \approx 8200$  K) is the most distinctive: it combines a strong infrared excess ( $\chi_{W2} \approx 50$ , fitted covering fraction  $\tau \approx 0.3$ ) with both an energy-balance match and significant mid-infrared variability in the NEOWISE epochs, the last indicating a *changing* rather than static excess. It is exactly the kind of multiply-anomalous target the framework is designed to surface and heads the follow-up list (its full ultraviolet-to-infrared SED, with the other top candidates, is shown in Fig. 12) — subject, as always, to the caveat that each axis taken alone has mundane explanations (here most plausibly a variable dusty disk). The energy-balance plane (Fig. 11) shows the diagnostic directly: the great majority of infrared-excess white dwarfs add infrared flux without any compensating ultraviolet deficit and so fall far from the one-to-one reprocessing line, exactly as expected for warm dust or a sub-stellar companion; only an object that both suppresses ultraviolet and re-emits in the infrared in matched proportion approaches the diagonal.

We stress that a multi-axis flag remains a prioritisation, not a detection. The ultraviolet axes in particular rest on extrapolating the photosphere across a wide wavelength baseline and on the shallow AIS depth, and a genuine energy-balance coincidence requires a deep, simultaneous ultraviolet measurement to confirm. The value of the framework is that it converts a one-dimensional infrared-excess list, where the candidate count is dominated by mundane confounders, into a ranked list whose top entries must satisfy several independent and physically distinct conditions at once — a far more stringent filter, and one that sharpens rather than inflates the follow-up target list.

## 7 Discussion

**The empirical limit and the forecast gap.** Our empirical 95% occurrence-rate limit,  $f < 2.4 \times 10^{-2}$  over  $N_{\text{eff}} = 7409$  searched white dwarfs, is nearly two orders of magnitude weaker

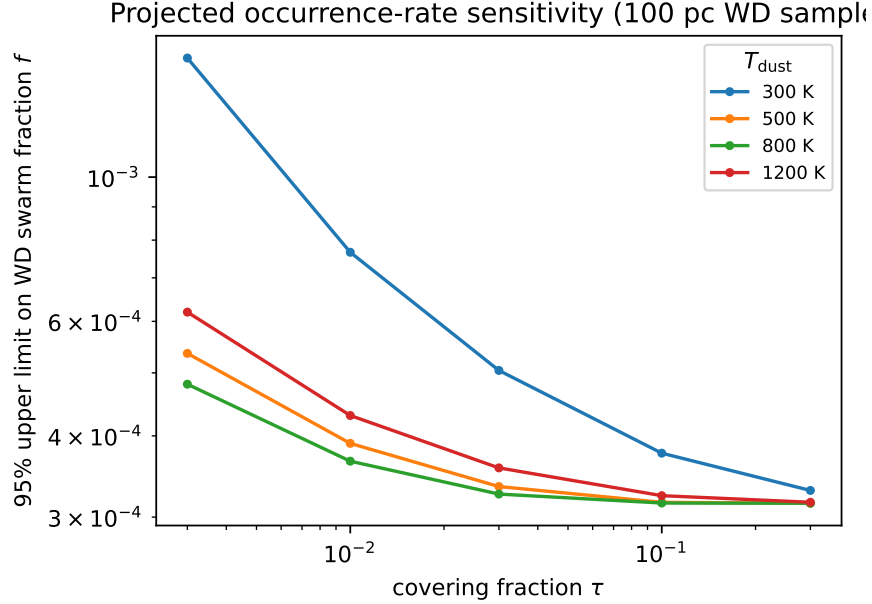


Figure 5: Projected 95% occurrence-rate upper limit on white-dwarf Dyson-swarm-like infrared excess as a function of covering fraction  $\tau$ , for several waste-heat temperatures  $T_{\text{dust}}$ , assuming a null search of the 100 pc sample.

than the idealised forecast ( $f < 3.2 \times 10^{-4}$ ; Section 5). It is essential to be clear about *why*, because the dominant factor is not the one a casual reading would suggest. The limit is set by the 156 cool-excess candidates that survive the funnel: the occurrence bound counts each as potential signal, so  $f \approx 156/N_{\text{eff}}$ . Were the search a clean null over the same sample, the bound would be  $\approx 3/N_{\text{eff}} \sim 4 \times 10^{-4}$  — comparable to the forecast. In other words, the gap is dominated by *residual candidate contamination*, not by the searchable fraction: two-band infrared excess simply cannot reject the cool-excess population (overwhelmingly brown-dwarf companions and warm dust), so they remain on the books as candidates and weaken the limit.

This is precisely the problem the multi-modal framework addresses. Requiring coincidence across independent axes collapses the 156 single-axis candidates to 16 flagged on two or more axes and a single object on three; an occurrence limit conditioned on the multi-axis criterion is therefore far tighter, at the cost of a completeness penalty (an object must be measurable on, and anomalous in, more than one axis) that we do not yet model fully. A secondary contributor to the gap is the searchable fraction itself: the forecast assumes every WISE-detected white dwarf is searched (9508), whereas the photospheric anchor requires a 2MASS near-infrared detection or, failing that, a Gaia BP/RP measurement, and Gaia-band anchoring carries a larger systematic because the optical white-dwarf SED departs from a blackbody (Balmer features, and for hot white dwarfs the optical lies near the spectral peak rather than on the Rayleigh–Jeans tail); we accordingly flag the anchor tier per source. Recovering the full idealised sensitivity therefore needs both deeper near-infrared photometry to restore the anchor (VHS, UKIDSS, or Euclid) and the multi-axis vetting to clear the candidate list. We regard  $f \sim 10^{-2}$ , candidate-limited, as the sensitivity of a present-day two-band search, and the multi-modal extension as the route back towards  $3.2 \times 10^{-4}$ .

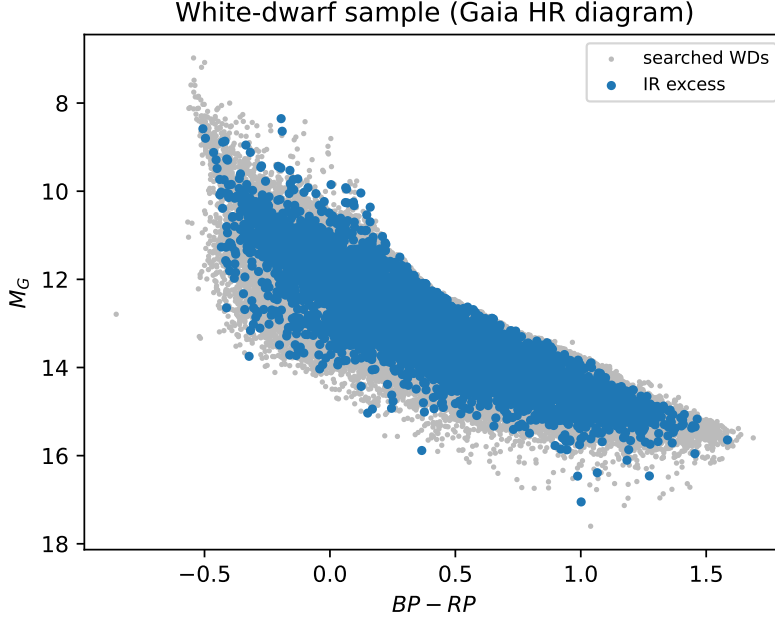


Figure 6: Gaia colour-magnitude diagram of the searched 300 pc white-dwarf sample (grey), the infrared-excess sources (blue) and the cool-excess candidates (red stars). Real data.

**Brown-dwarf companions: the dominant astrophysical false positive.** The candidates are cool-excess sources, and the leading natural explanation for cool infrared excess around a white dwarf — ahead even of circumstellar dust — is an unresolved substellar (brown-dwarf) companion, whose photosphere adds flux longward of  $\sim 2\mu\text{m}$  [Farihi et al., 2005]. White dwarf–brown dwarf binaries occur at the  $\sim 0.5\text{--}2\%$  level, so across 7 409 searched white dwarfs of order tens of unresolved companions are expected — a yield comparable to or exceeding our 156 candidates. A brown-dwarf companion is in principle separable from a Dyson swarm by its distinctive  $W1 - W2$  colour and by flux already present in the  $K$  band, but two-band photometry cannot make that separation alone. The candidate list is therefore most conservatively read as a small sample of cool-excess white dwarfs for spectroscopic or multi-band follow-up, the large majority of which will prove to be brown-dwarf companions or cool dust rather than technosignatures.

**The dust/structure degeneracy and the two-band fit.** A partial Dyson swarm and warm circumstellar dust produce indistinguishable  $W1/W2$  excess. Moreover,  $(T_{\text{dust}}, \tau)$  is fit from only the  $W1$  and  $W2$  excess: two parameters from two bands, so the inferred temperatures (and hence the “ $T_{\text{dust}} \sim 500\text{ K}$ ” classification) carry large, correlated uncertainties, broken only weakly by the 2MASS and  $W3/W4$  upper limits. We do not claim any candidate is a technosignature; the discriminator prioritises rather than confirms. Optical variability from ZTF (dust transits as in WD 1145+017 favour a natural disk; a smooth, stable excess does not) and the full  $J$ -through- $W4$  SED are the degeneracy-breakers and are built into the pipeline as shortlist follow-up.

**Limitations.** In summary, the present empirical search is limited by (i) the searchable fraction, set by the near-infrared/Gaia SED anchor; (ii) the larger systematic of Gaia-anchored predictions; (iii) interstellar reddening, which, although modest over the 300 pc volume ( $E(B - V) \lesssim 0.02$  within 100 pc, of order 0.05–0.1 towards the more distant sightlines), reddens the anchor, grows with dis-

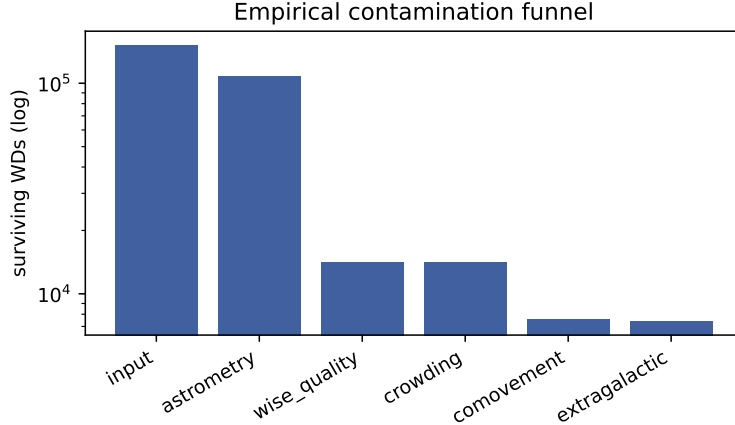


Figure 7: Empirical contamination funnel on the real 300 pc data: white dwarfs surviving each successive cut (log scale). The WISE-quality and co-movement cuts dominate the attrition.

tance and is a sub-dominant contributor to the systematic floor; (iv) the dust/swarm/companion degeneracy of two-band photometry; and (v) the residual chance-alignment and brown-dwarf contamination budgeted above, which is comparable to the candidate count. None of these preclude the limit, but they set its interpretation.

**Beyond infrared excess: the multi-modal expansion.** The single most important lesson of the empirical search is that two-band infrared excess, taken alone, cannot distinguish a technosignature from the dominant astrophysical false positives, and that the candidate count is set by those false positives rather than by any genuine signal. The multi-modal framework (Sections 3.5, 6.1) is our response: by requiring coincidence across physically distinct and independently-failing axes — an ultraviolet deficit, an energy-balance match, a significant optical period, and optical and infrared variability — it suppresses exactly the confounders that inflate a one-dimensional list. The energy-balance axis is the most discriminating in principle, because the matched-absorption/re-emission coincidence it tests is the defining thermodynamic property of a reprocessing structure and is not reproduced by dust, by a companion, or by a chance alignment. In practice its reach is presently limited by the depth of all-sky ultraviolet photometry: GALEX AIS detects only the hotter white dwarfs, so the axis is available for a minority of the sample, and a definitive test of any energy-balanced candidate needs a deep, contemporaneous ultraviolet measurement (e.g. from *Swift*/UVOT or a future UV facility such as *UVEX* or *ULTRASAT*). The framework is built so that each additional axis tightens the ranking without changing the core methodology, and a candidate surviving several independent axes would be a far stronger anomaly than any infrared-excess outlier.

**Outlook.** Three advances would materially strengthen the search. Longer-wavelength photometry — WISE *W3/W4* for the brightest candidates, and JWST/MIRI for the rest — directly measures the excess temperature and breaks the dust/swarm/companion degeneracy that two-band data cannot. Time-domain optical photometry from ZTF, and from Rubin/LSST once operational, tests for the dust-transit variability that marks an active debris disk. And deeper or larger samples — the full Gaia DR4 white-dwarf catalogue beyond the 300 pc volume searched here, together with deep near-infrared surveys to restore the SED anchor — would push  $N_{\text{eff}}$  toward the forecast

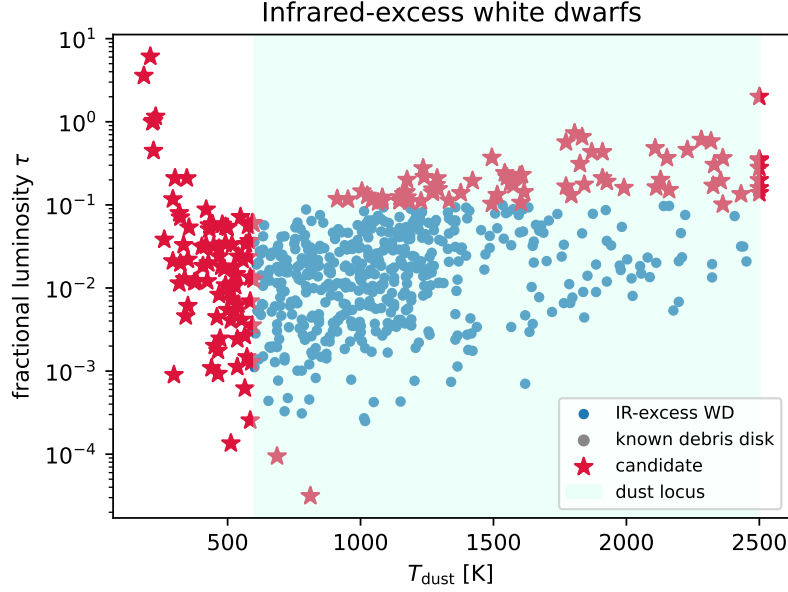


Figure 8: Fitted  $(T_{\text{dust}}, \tau)$  of the infrared-excess white dwarfs (blue), known debris disks (grey) and candidates (red stars), with the dust locus shaded. Cool ( $T_{\text{dust}} \lesssim 600$  K) excesses fall outside the locus. Real data;  $(T_{\text{dust}}, \tau)$  are fit from  $W1/W2$  only and carry large correlated uncertainties (Section 7).

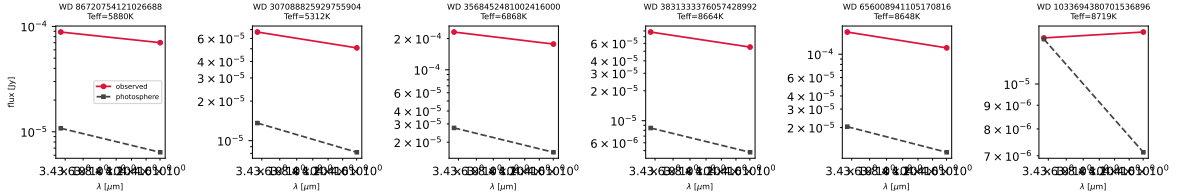


Figure 9: Observed (red) versus predicted-photosphere (grey) spectral energy distributions of the cool-excess candidates, showing the infrared excess in  $W1/W2$ . Real data.

value; combined with the multi-axis vetting that clears the candidate list, this is the route from the present candidate-limited  $f \sim 10^{-2}$  toward the idealised  $\sim 10^{-4}$ . The released pipeline is built to ingest each of these with no change to the core methodology.

## 8 Conclusions

We have designed, implemented, validated, and carried out the first systematic search for infrared-excess technosignatures around white dwarfs. The methodology is contamination-robust by construction: its novel Gaia–WISE co-movement cut directly addresses the background-contamination failure mode of earlier broadband searches, removing roughly half of all chance-aligned infrared sources overall and essentially all of them for the fast-moving majority. Applied to the real 300 pc white-dwarf sample, the search vets 7 409 white dwarfs, recovers 692 with infrared excess (the warm debris-disk population), and places an empirical 95% occurrence-rate limit of  $f < 2.4 \times 10^{-2}$  on warm, swarm-like excess, with 156 cool-excess candidates — most likely brown-dwarf companions

Table 3: Cool-excess candidates from the empirical 300 pc search (outside the dust locus; 156 in total, the 20 highest-ranked shown).  $(T_{\text{dust}}, \tau)$  are fit from  $W1/W2$  only and are indicative. Auto-generated from the pipeline.

| Gaia EDR3 source_id | RA        | Dec       | $T_{\text{eff}}$ | $T_{\text{dust}}$ | $\tau$ | $\chi_{W2}$ |
|---------------------|-----------|-----------|------------------|-------------------|--------|-------------|
| 86720754121026688   | 35.14765  | 18.54212  | 5880             | 1773              | 0.567  | 13.5        |
| 307088825929755904  | 19.01818  | 27.91262  | 5312             | 1870              | 0.433  | 10.6        |
| 3568452481002416000 | 178.83549 | -16.42336 | 6868             | 1910              | 0.433  | 31.5        |
| 3831333376057428992 | 156.41598 | 0.38730   | 8664             | 2363              | 0.366  | 12.8        |
| 656008941105170816  | 125.85147 | 16.80389  | 8648             | 2330              | 0.304  | 19.8        |
| 1033694380701536896 | 127.59838 | 55.52525  | 8719             | 221               | 1.026  | 4.7         |
| 5216873096850721280 | 135.51689 | -74.90368 | 9072             | 1833              | 0.656  | 29.3        |
| 6773854228332228608 | 314.87888 | -39.95837 | 7267             | 2153              | 0.367  | 11.9        |
| 6541715644029638912 | 347.03072 | -43.23834 | 6697             | 2229              | 0.459  | 13.8        |
| 6330824366020176768 | 220.85058 | -6.72306  | 8473             | 2500              | 2.004  | 39.3        |
| 1289022219283937408 | 226.45546 | 33.06923  | 4285             | 1805              | 0.729  | 21.9        |
| 1889057183787160448 | 339.25085 | 31.30970  | 8826             | 2500              | 0.354  | 22.6        |
| 3957924620825272192 | 191.54940 | 24.90177  | 7259             | 2108              | 0.483  | 24.7        |
| 1528971665472343040 | 192.06717 | 44.92839  | 7354             | 229               | 1.153  | 4.7         |
| 3934459045528378368 | 192.68486 | 15.83242  | 8211             | 2500              | 0.341  | 53.0        |
| 1557096756297688704 | 198.52820 | 52.39745  | 11746            | 221               | 0.448  | 4.0         |
| 1661896951912448512 | 204.41223 | 58.43782  | 6857             | 2318              | 0.583  | 15.7        |
| 3932931205402346496 | 187.47139 | 14.12687  | 10575            | 217               | 0.991  | 5.4         |
| 5093513385476212224 | 64.54309  | -17.56488 | 5532             | 209               | 6.098  | 4.4         |
| 235890564653455488  | 50.68951  | 39.07923  | 6267             | 2282              | 0.607  | 21.2        |

or cool dust — remaining for follow-up. Because two-band infrared excess cannot by itself separate a technosignature from these confounders, we extend the search into a multi-modal anomaly framework that scores each white dwarf on several independent and physically distinct axes — an ultraviolet deficit, an energy-balance match between ultraviolet absorption and infrared re-emission, a significant optical period, and optical and mid-infrared variability — and promotes only the 16 objects flagged on two or more axes at once, one of which is flagged on three. This converts a confounder-dominated candidate list into a sharply ranked one and is the principal lever for tightening the candidate-limited occurrence bound. A calibrated forecast shows that deeper near-infrared anchoring would further tighten the limit toward  $3.2 \times 10^{-4}$ . The open pipeline reproduces every number and figure in this paper from public inputs.

## Data and code availability

The complete, reproducible pipeline is openly available (Zenodo DOI [\[1\]](#)). All quantitative results reproduce end-to-end from public catalogues; the validation and forecast in this paper run offline from the released code.

## Acknowledgements

This work made use of Gaia, WISE/CatWISE2020/unWISE, 2MASS, and the VizieR and CDS X-Match services.

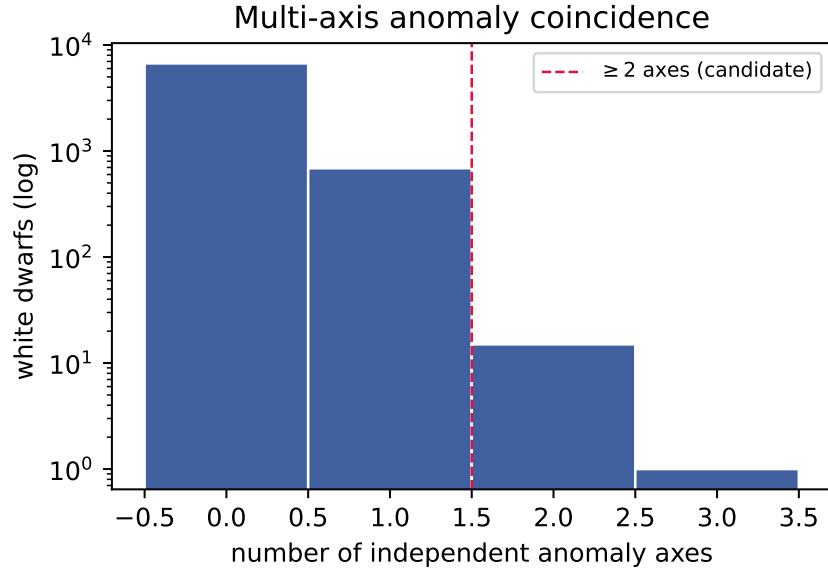


Figure 10: Distribution of the number of independent anomaly axes per white dwarf in the clean 300 pc sample. Objects to the right of the dashed line ( $\geq 2$  axes) are the multi-modal candidates. Real data.

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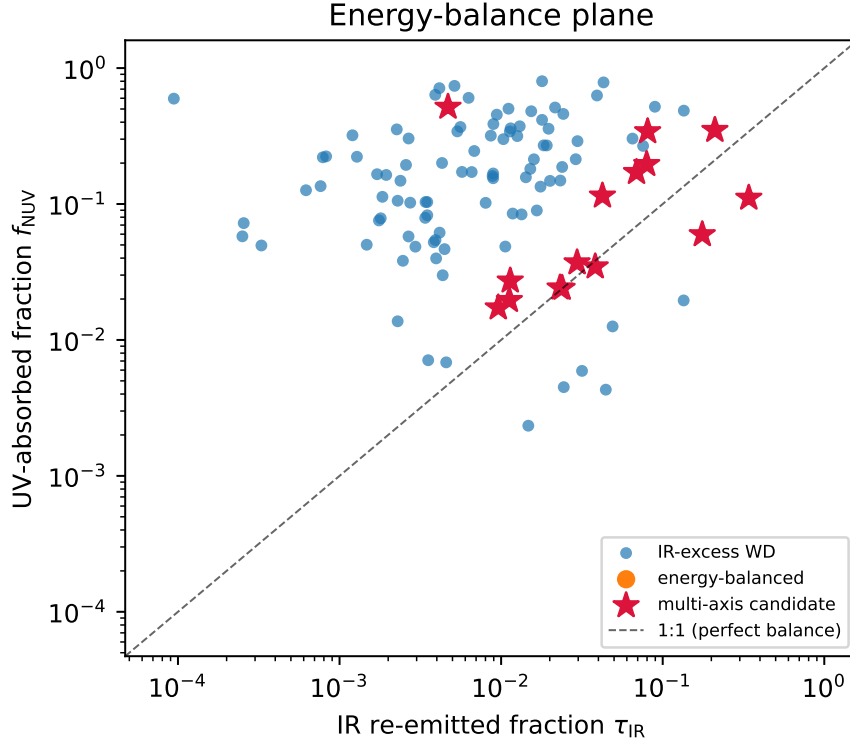


Figure 11: Energy-balance plane: ultraviolet-absorbed fraction  $f_{\text{NUV}}$  versus infrared re-emitted fraction  $\tau_{\text{IR}}$  for white dwarfs with a GALEX NUV measurement. The dashed line is perfect reprocessing ( $f_{\text{NUV}} = \tau_{\text{IR}}$ ). Warm dust and brown-dwarf companions add infrared flux without removing ultraviolet flux and fall to the lower right; only a structure that intercepts and re-radiates starlight approaches the diagonal. Real data.

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*Multi-axis candidate SED panel — regenerated by the pipeline.*

Figure 12: Full ultraviolet-to-infrared spectral energy distributions of the highest-ranked multi-axis candidates: observed flux (red) against the blackbody photosphere (grey) fitted from the Gaia solid angle and  $T_{\text{eff}}$ . For an energy-balanced candidate the diagnostic is visible in a single panel — the GALEX ultraviolet points fall below the photosphere while the  $W1/W2$  points rise above it. Real data.

Table 4: Multi-axis candidates from the 300 pc search (flagged on  $\geq 2$  independent evidence axes), ranked by the aggregate multi-modal score; up to 20 shown. Auto-generated from the pipeline.

| Gaia EDR3 source_id | $T_{\text{eff}}$ | $n_{\text{axes}}$ | axes                                    | score |
|---------------------|------------------|-------------------|-----------------------------------------|-------|
| 3934459045528378368 | 8211             | 3                 | ir_excess,energy_balance,ir_variability | 0.57  |
| 1037258898615762048 | 15416            | 2                 | ir_excess,ir_variability                | 0.43  |
| 1884586638232948096 | 9984             | 2                 | ir_excess,energy_balance                | 0.37  |
| 2575516721465083904 | 11116            | 2                 | ir_excess,energy_balance                | 0.37  |
| 589660732992427392  | 11695            | 2                 | ir_excess,energy_balance                | 0.37  |
| 2761950707640652288 | 15278            | 2                 | ir_excess,energy_balance                | 0.37  |
| 6699657603142163456 | 7927             | 2                 | ir_excess,energy_balance                | 0.36  |
| 4187838721315035264 | 16496            | 2                 | ir_excess,energy_balance                | 0.35  |
| 3202394239430450304 | 13677            | 2                 | ir_excess,energy_balance                | 0.35  |
| 405084650886316160  | 10274            | 2                 | ir_excess,energy_balance                | 0.31  |
| 1698019478002207872 | 10447            | 2                 | ir_excess,energy_balance                | 0.31  |
| 1121123185951605632 | 16672            | 2                 | ir_excess,optical_variability           | 0.31  |
| 1773969996038337792 | 6896             | 2                 | ir_excess,energy_balance                | 0.29  |
| 1520345099103040256 | 7122             | 2                 | ir_excess,energy_balance                | 0.28  |
| 3527241994919570688 | 16516            | 2                 | ir_excess,energy_balance                | 0.25  |
| 1242732500581829632 | 22939            | 2                 | ir_excess,uv_deficit                    | 0.17  |

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